



C23-EE-405

**23439**

**BOARD DIPLOMA EXAMINATION, (C-23)**

**MARCH/APRIL—2025**

**DEEE – FOURTH SEMESTER EXAMINATION**

**GENERAL MECHANICAL ENGINEERING**

Time : 3 hours ]

[ Total Marks : 80

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**PART—A**

3×10=30

- Instructions :**
- (1) Answer **all** questions.
  - (2) Each question carries **three** marks.
  - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. Define the terms (a) tensile stress and (b) compressive stress.
- 2. Draw stress-strain diagram for a ductile material and show salient points in it.
- 3. What is a 'Shaft' and how are they classified.
- 4. Write the formula for the polar moment of inertia of
  - (a) a circular solid shaft of diameter 'd'
  - (b) a hollow shaft of internal diameter ' $d_i$ ' and external diameter ' $d_o$ '.
- 5. Compare four-stroke and two-stroke IC engines.
- 6. List out any six components of IC engine.
- 7. Compare fire tube and water tube boilers.
- 8. State the working principle of a steam turbine.
- 9. What is priming in a centrifugal pump?
- 10. What are the applications of a reciprocating pump?

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## PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.  
(2) Each question carries **ten** marks.  
(3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

11. A load of 150 kN was applied over a specimen of 20 mm diameter and 150 mm length in a compression testing machine. The reduction in length due to compressive force was 0.2 mm. Determine the Young's modulus of elasticity.
12. Calculate the power transmitted by a shaft of 200 mm diameter running at 300 rpm, if the shear stress in the shaft is not to exceed 75 N/mm<sup>2</sup>.
13. Explain the working principle of a 4-stroke diesel engine with a neat sketch.
14. Compare petrol and diesel engines.
15. Explain the working of a 'Lamont boiler' with a neat sketch.
16. Explain the working of Francis turbine with a neat sketch.
17. Draw a neat sketch of a centrifugal pump and explain its working.
18. How a single acting reciprocating pump works? Explain with the help of a neat sketch.

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